MS-DCAFNET: MULTI-SCALE DAMAGE-CONDITIONED ADAPTIVE FUSION NETWORK FOR BUILDING DAMAGE ASSESSMENT FROM BI-TEMPORAL SATELLITE IMAGERY

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DECLARATION

I, Hiba V S (Reg. No. IES23CD008), IES College of Engineering, Thrissur, hereby declare that the internship report entitled **"MS-DCAFNet: Multi-Scale Damage-Conditioned Adaptive Fusion Network for Building Damage Assessment from Bi-Temporal Satellite Imagery"** submitted to the Indian Institute of Information Technology Kottayam towards the partial fulfilment of the requirements of the Summer Internship Programme at the FACTS-H Lab is an original work carried out by me under the guidance of Dr. Ansith S. The work presented in this report has not been submitted elsewhere for the award of any degree or diploma. Wherever the work of others has been used, proper acknowledgement has been made.

Kottayam-686635 Signature

July 2026 Hiba V S

CERTIFICATE

This is to certify that the internship work entitled **"MS-DCAFNet: Multi-Scale Damage-Conditioned Adaptive Fusion Network for Building Damage Assessment from Bi-Temporal Satellite Imagery"** carried out by **Hiba V S** under my guidance has been completed satisfactorily during the internship period at the Indian Institute of Information Technology Kottayam.

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ABSTRACT

Building Damage Assessment (BDA) from bi-temporal satellite imagery plays a critical role in rapid disaster response, but existing approaches struggle to preserve fine structural detail during feature extraction and to model complex temporal changes across multiple spatial scales. This report presents the work carried out during a summer research internship at the FACTS-H Lab, IIIT Kottayam, under the guidance of Dr. Ansith S, towards the development of MS-DCAFNet (Multi-Scale Damage-Conditioned Adaptive Fusion Network) - a Siamese deep learning architecture for pixel-level, five-class building damage classification from pre- and post-disaster satellite image pairs.

The internship work involved designing a wavelet-guided multi-scale encoder that uses learnable Haar decomposition in place of conventional downsampling to retain edge and structural information, and a Hierarchical Damage-Conditioned Attention Fusion (H-DCAF) module that adaptively combines pre-disaster, post-disaster, and change-sensitive features at every encoder scale. A UNet++-style decoder with deep supervision was implemented to recover full-resolution damage maps, trained using a composite loss combining focal, dice, and boundary-aware terms to address the severe class imbalance present in the dataset. In addition, a comprehensive explainability (XAI) pipeline was built around the trained model, combining architecture-native attention, gradient-based attribution, perturbation-based methods, and relevance propagation, along with quantitative faithfulness evaluation.

The model was trained and evaluated on the xBD benchmark dataset under a strict event-based data partitioning protocol to prevent spatial leakage between training and test events. The final model achieved an overall accuracy of 0.9403, a mean Intersection-over-Union of 0.4459, and a Cohen's Kappa of 0.6117, with ablation experiments confirming the effectiveness of the proposed H-DCAF fusion strategy over conventional temporal fusion methods. This report documents the methodology, implementation, experimental results, and interpretability analysis carried out over the course of the internship.

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Chapter 1

Introduction

1.1 Background

Natural disasters such as earthquakes, floods, hurricanes, wildfires, and tsunamis cause extensive damage to buildings and critical infrastructure every year. Rapid and accurate Building Damage Assessment (BDA) is essential for emergency response, resource allocation, and post-disaster recovery planning. Traditional field surveys, while detailed, are slow, labour-intensive, and often difficult to carry out immediately after a disaster due to accessibility constraints and safety risks. High-resolution satellite imagery, combined with recent advances in deep learning, offers an efficient alternative for large-area, rapid damage assessment.

Publicly available benchmarks such as the xBD dataset have enabled the development of automated BDA systems using Convolutional Neural Networks (CNNs), Siamese architectures, attention mechanisms, and transformer-based models. These methods have made steady progress in building localisation, damage classification, and change detection from bi-temporal image pairs. However, two key challenges remain largely unresolved: (i) conventional downsampling operations such as max-pooling and strided convolution tend to discard high-frequency structural information — such as building edges and roof geometry — that is important for distinguishing between damage severities, and (ii) commonly used temporal fusion strategies such as subtraction, differencing, and concatenation are often too simplistic to capture the complex, multi-scale structural changes that occur between pre- and post-disaster imagery.

1.2 Motivation

This internship was undertaken at the FACTS-H Lab, IIIT Kottayam, under the guidance of Dr. Ansith S, with the objective of designing a deep learning framework that directly addresses these limitations. The motivation was twofold: first, to build a more structurally aware encoder that preserves fine-grained detail during feature extraction, and second, to design a fusion mechanism that can adaptively combine temporal information at multiple spatial scales rather than relying on a single, fixed fusion strategy at the bottleneck. An additional focus of the internship was interpretability — given that BDA models are intended to support safety-critical, real-world disaster response decisions, it was considered important to be able to explain and verify why the model makes the predictions it does, rather than treating it purely as a black box.

1.3 Objectives

The key objectives of the internship were as follows:

1. To design and implement a wavelet-guided multi-scale Siamese encoder that preserves structural detail using learnable Haar wavelet decomposition instead of conventional downsampling.

2. To design a Hierarchical Damage-Conditioned Attention Fusion (H-DCAF) module that adaptively fuses pre-disaster, post-disaster, and change-sensitive features across multiple encoder scales.

3. To implement a decoder with deep supervision, and a composite loss function suited to the severe class imbalance present in the xBD dataset.

4. To train and rigorously evaluate the proposed model on the xBD benchmark using an event-based data partitioning protocol that prevents spatial leakage between training and test sets.

5. To build a comprehensive explainability (XAI) pipeline for the trained model, combining multiple attribution methods with quantitative faithfulness evaluation.

6. To carry out ablation experiments to isolate and validate the individual contribution of the proposed H-DCAF fusion module.

1.4 Scope

The scope of the internship was restricted to the design, implementation, training, and evaluation of the MS-DCAFNet architecture on the xBD dataset for five-class pixel-level damage segmentation (Background, No Damage, Minor Damage, Major Damage, Destroyed), along with the accompanying explainability analysis. Deployment-level engineering concerns, such as real-time inference optimisation or integration into an operational disaster-response pipeline, were outside the scope of this internship, though they are discussed briefly as directions for future work.

Chapter 2

Literature Review

2.1 CNN-Based Approaches to Building Damage Assessment

Early deep learning approaches to BDA were built on CNN-based encoder-decoder architectures. U-Net and its variants demonstrated strong performance for building localisation and damage mapping [2], [13]. RescueNet extended this line of work by combining building segmentation with joint damage assessment directly from satellite imagery [3]. However, many of these early methods offered limited modelling of the temporal relationship between pre-disaster and post-disaster observations, treating damage assessment largely as a single-image segmentation problem rather than a genuine change-detection task.

2.2 Bi-Temporal and Siamese Architectures

Later work moved toward explicitly bi-temporal architectures that directly compare pre- and post-disaster imagery. ChangeOS introduced object-based semantic change detection for damage assessment [4], while BDANet employed Siamese feature extraction for post-disaster analysis from ultra-high-resolution imagery [14]. Cross-directional feature fusion and Siamese transformer frameworks further improved temporal representation learning [7], [10]. Despite these advances, subtraction, differencing, and simple concatenation remain the dominant fusion strategies used by most existing methods — an observation that directly motivated the design of the H-DCAF module in this work.

2.3 Attention Mechanisms and Transformer-Based Models

Attention mechanisms and transformer architectures have further improved contextual modelling and long-range dependency capture in BDA. The Convolutional Block Attention Module (CBAM) introduced lightweight channel and spatial attention for feature refinement [5]. Transformer-enhanced Siamese architectures such as SLgViT and DamageCAT have shown improved capability for change detection and damage categorisation from remote sensing imagery [6], [15], [16]. However, these approaches largely continue to rely on conventional feature extraction pipelines that are susceptible to the loss of fine spatial detail during downsampling.

2.4 Frequency-Aware Feature Learning

Frequency-domain learning has received growing attention as a way to preserve structural detail during feature extraction. Wavelet pooling demonstrated the benefits of frequency decomposition for retaining structural information in CNNs [8], while the High-Frequency Attention Network (HFA-Net) incorporated high-frequency attention specifically for building change detection from very high-resolution imagery [9]. Broader remote sensing surveys have similarly highlighted the importance of multi-scale representation learning for change detection tasks [17]. However, the combination of frequency-aware feature extraction with adaptive temporal fusion for multi-class BDA remained largely unexplored prior to this work, representing a clear gap that the proposed wavelet-guided encoder addresses.

2.5 Explainability in Remote Sensing

Interpretability has become an increasingly important consideration for deep learning models used in remote sensing, particularly for safety-critical applications such as disaster response. Prior work has investigated attention-based explainability in remote sensing analysis [11], while general-purpose attribution techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM), Integrated Gradients, Layer-wise Relevance Propagation, and LIME have been widely used for model interpretation and feature attribution across computer vision tasks [12], [18]–[20]. Despite this, comprehensive, multi-method interpretability analysis specifically for multi-class BDA remains limited, motivating the explainability pipeline developed as part of this internship.

2.6 Summary

Overall, the reviewed literature reveals two persistent gaps in existing BDA approaches: the loss of fine-grained structural information during conventional feature extraction, and the limited expressiveness of simple temporal fusion strategies for modelling multi-scale structural change. These two gaps directly motivate the two core technical contributions of this internship project — the wavelet-guided multi-scale encoder and the Hierarchical Damage-Conditioned Attention Fusion (H-DCAF) module — described in detail in Chapter 3.

Chapter 3

Methodology

3.1 Overview

The proposed MS-DCAFNet framework maps a co-registered bi-temporal image pair (I^{pre}, I^{post}) to a dense five-class damage map. It consists of four components: (i) a weight-shared Wavelet-Guided Multi-Scale Siamese Encoder that extracts four-scale feature pyramids independently from each temporal image, (ii) a Hierarchical Damage-Conditioned Attention Fusion module (H-DCAF) that fuses pre-/post-disaster features at every encoder scale and adaptively aggregates them into a single bottleneck representation, (iii) a UNet++-style decoder with deep supervision that progressively restores spatial resolution using the H-DCAF fused features as skip connections, and (iv) a composite loss function that jointly supervises the main and two auxiliary prediction heads. The overall architecture is illustrated in Figure 3.1.

3.2 Wavelet-Guided Multi-Scale Siamese Encoder

Both temporal branches (pre-disaster and post-disaster) share identical encoder weights, so that any divergence between the two streams reflects genuine visual change rather than parameter asymmetry. The encoder produces four-scale feature maps with channel widths $[64, 128, 256, 512]$ at spatial resolutions $[H, H/2, H/4, H/8]$.

3.2.1 Double-Convolution Block

Every encoder stage and decoder up-block reuses the same double-convolution primitive: two successive 3×3 convolutions, each followed by batch normalisation and a ReLU activation, preserving spatial dimensions via same-padding.

3.2.2 Learnable Haar Wavelet Downsampling

Rather than using conventional max-pooling or strided convolution — both of which discard high-frequency edge information — spatial resolution is halved using a *learnable* Haar wavelet decomposition. Each 2×2 neighbourhood is decomposed into four polyphase sub-bands: LL (low-low, structural), LH (horizontal detail), HL (vertical detail), and HH (diagonal detail). A learnable band-weighting vector, normalised via softmax, weights these four sub-bands; the

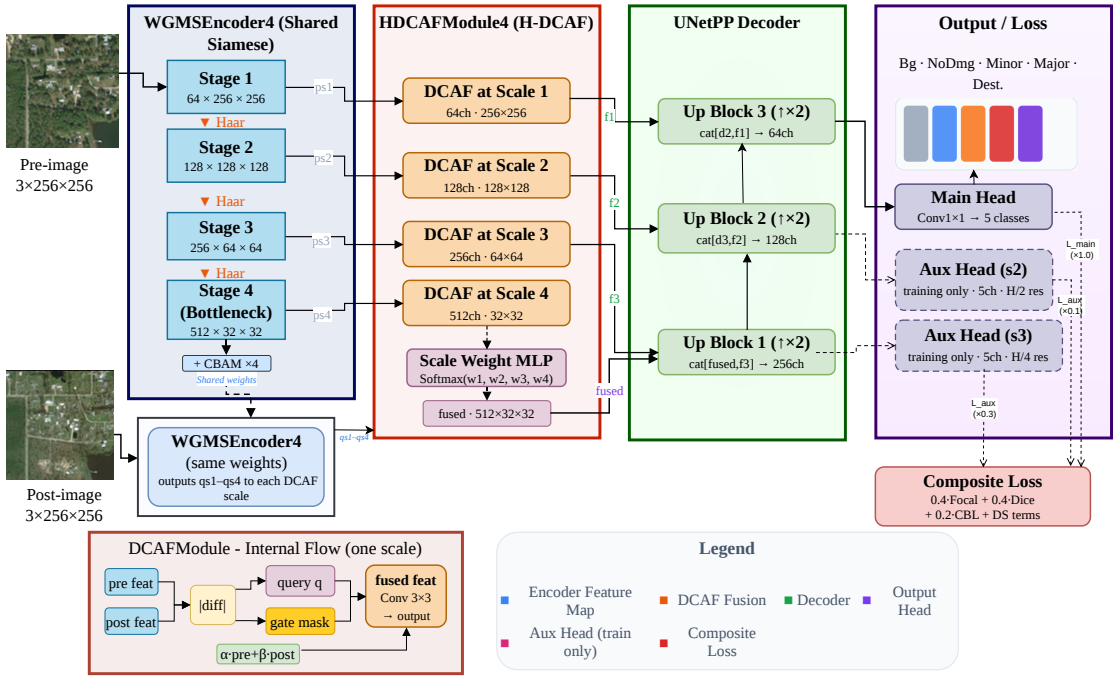


Figure 3.1: Overview of the MS-DCAFNet architecture: the Wavelet-Guided Multi-Scale Siamese Encoder (WGMSEncoder4), the Hierarchical Damage-Conditioned Attention Fusion module (H-DCAF), the UNet++ decoder, and the composite loss / output heads. The inset shows the internal flow of a single DCAF module at one encoder scale.

weighted sub-bands are concatenated with an unweighted residual copy of LL to preserve low-frequency content even if the network learns to suppress it, and the result is projected back to the target channel width via a 1×1 convolution.

3.2.3 Convolutional Block Attention Module (CBAM)

A CBAM module recalibrates features along the channel and spatial dimensions after every double-convolution block. Channel attention is computed using a shared bottleneck MLP applied to both the global-average-pooled and global-max-pooled descriptors of the input; spatial attention is computed via a 7×7 convolution over channel-pooled descriptors. One CBAM module is instantiated independently at each of the four encoder stages.

3.3 Hierarchical Damage-Conditioned Attention Fusion (H-DCAF)

Rather than fusing the temporal streams only at the network bottleneck — the conventional approach in most Siamese BDA architectures — H-DCAF performs an independent, context-aware fusion at *all four* encoder scales and then adaptively aggregates the four fused representations using a learned, change-magnitude-driven weighting scheme.

3.3.1 DCAF Module at a Single Scale

At each scale, the Damage-Conditioned Attention Fusion (DCAF) sub-module receives the pre- and post-disaster feature maps and produces a fused output in four steps:

1. **Absolute-difference change map** — the element-wise absolute difference between post- and pre-disaster features is computed as a first-order change signal.

2. **Global change query** — the change map is globally average-pooled and passed through a two-layer bottleneck MLP with reduction ratio 8, producing a global change-query vector that is broadcast spatially.

3. **Change-gated spatial mask** — the query-modulated pre-disaster context, the raw post-disaster context, and the change magnitude are concatenated and passed through a two-layer convolutional gate (with an intermediate 1×1 convolution followed by a 3×3 convolution and sigmoid activation), producing a spatial gate mask.

4. **Soft temporal gating and fusion** — two 1×1 convolutional projections produce per-channel scores for the pre- and post-disaster streams, jointly normalised via softmax so that their combination weights sum to one. The resulting temporally-blended feature is modulated by the gate mask and passed through a final 3×3 convolution to produce the fused output at that scale.

3.3.2 Learnable Scale-Weighting Aggregation

The four per-scale fused features carry damage evidence of different granularity — fine structural cracks at the finest scale versus block-level collapse at the coarsest scale. A Scale-Weight MLP learns content-adaptive aggregation weights driven by the observed change magnitude at each scale: the per-scale change descriptors are concatenated and passed through a two-layer MLP with a softmax output, yielding four scalar weights that sum to one. The three finer-scale fused features are projected to the bottleneck channel width and adaptively pooled to the bottleneck spatial resolution before being combined with the bottleneck-scale feature using the

learned weights, followed by a final 3×3 convolution to produce the bottleneck representation passed to the decoder.

3.4 UNet++ Decoder with Deep Supervision

The decoder recovers full spatial resolution through three up-sampling blocks. Each block bilinearly resizes its input to match the spatial resolution of the corresponding DCAF skip feature (rather than the raw, temporally-uninformed encoder features), concatenates the two along the channel axis, and applies the double-convolution primitive. The main prediction head maps the final decoder output to five class logits at full resolution. Two auxiliary heads, active during training only, supervise the intermediate decoder stages at coarser resolution and are discarded at inference.

3.5 Composite Loss Function

The xBD dataset exhibits severe class imbalance — background and no-damage pixels routinely exceed 90% of a scene. To address this, the composite loss combines three complementary terms:

- **Focal loss with label smoothing** — a class-weighted, label-smoothed cross-entropy with a per-pixel focusing modulation ($\gamma = 2.0$) that down-weights easy, already-confident pixels and concentrates learning on hard examples.
- **Multi-class Dice loss** — optimises spatial overlap directly, averaged over both the batch and class dimensions, and is inherently robust to class imbalance.
- **Contrastive Boundary Loss (CBL)** — penalises errors at damage-class boundaries more heavily, using a Sobel-derived boundary indicator to identify boundary pixels and up-weighting the cross-entropy loss at those locations.

The three terms are combined as $0.4\,\mathcal{L}_{Focal} + 0.4\,\mathcal{L}_{Dice} + 0.2\,\mathcal{L}_{CBL}$, applied to all three prediction heads (main, and two auxiliary heads) with diminishing weights of 1.0, 0.3, and 0.1 respectively, reflecting their coarser spatial resolution.

3.6 Dataset and Pre-Processing

MS-DCAFNet is trained and evaluated on the xBD dataset, which provides co-registered pre- and post-disaster RGB image pairs across multiple disaster events worldwide with polygon-level building damage annotations on a four-point ordinal scale (No Damage, Minor Damage, Major Damage, Destroyed). Polygon annotations are rasterised to single-channel masks, and

each image triplet is resized to 256×256 (bilinear for images, nearest-neighbour for masks, to preserve discrete class boundaries). RGB channels are normalised using ImageNet statistics.

A critical methodological choice adopted for this work is **event-based data partitioning**: the training portion is partitioned event-wise, not image-wise, into training and validation subsets, with one disaster event held out entirely for validation. This prevents spatial leakage between geographically adjacent tiles of the same event, which would otherwise produce artificially inflated validation performance.

3.6.1 Data Augmentation

Augmentations are applied synchronously to the pre-image, post-image, and mask: horizontal/vertical flipping (probability 0.5 each), class-conditioned $90^\circ \times k$ rotation for patches containing minority damage classes (probability 0.7), and CutMix applied to mini-batches containing Minor- or Major-damage pixels (probability 0.5).

3.7 Training Protocol

The model is trained for up to 120 epochs with early stopping (patience of 30 epochs), monitored on the validation $F1_{damaged}$ score. Optimisation uses AdamW with weight decay 10^{-4}. The learning rate follows a linear warm-up phase for 5 epochs followed by cosine annealing to a minimum of 10^{-6} over 115 epochs, starting from a peak learning rate of 3×10^{-4}. Gradient norms are clipped to 1.0 after every backward pass. All experiments use $H = W = 256$ and batch size 16.

3.8 Explainability Analysis Pipeline

Given that BDA models are intended to support safety-critical disaster-response decisions, a post-hoc explainability (XAI) pipeline was built around the trained, frozen model, combining four complementary families of explanation methods with quantitative faithfulness evaluation.

3.8.1 Native Architectural Attention

Three sources of intrinsic attention require no backpropagation and add no computational overhead beyond the standard forward pass: the CBAM spatial attention maps at each encoder stage, the H-DCAF gate masks at each fusion scale, and the learned scale-weighting MLP outputs, which reveal which encoder scale the model relies on most for a given prediction.

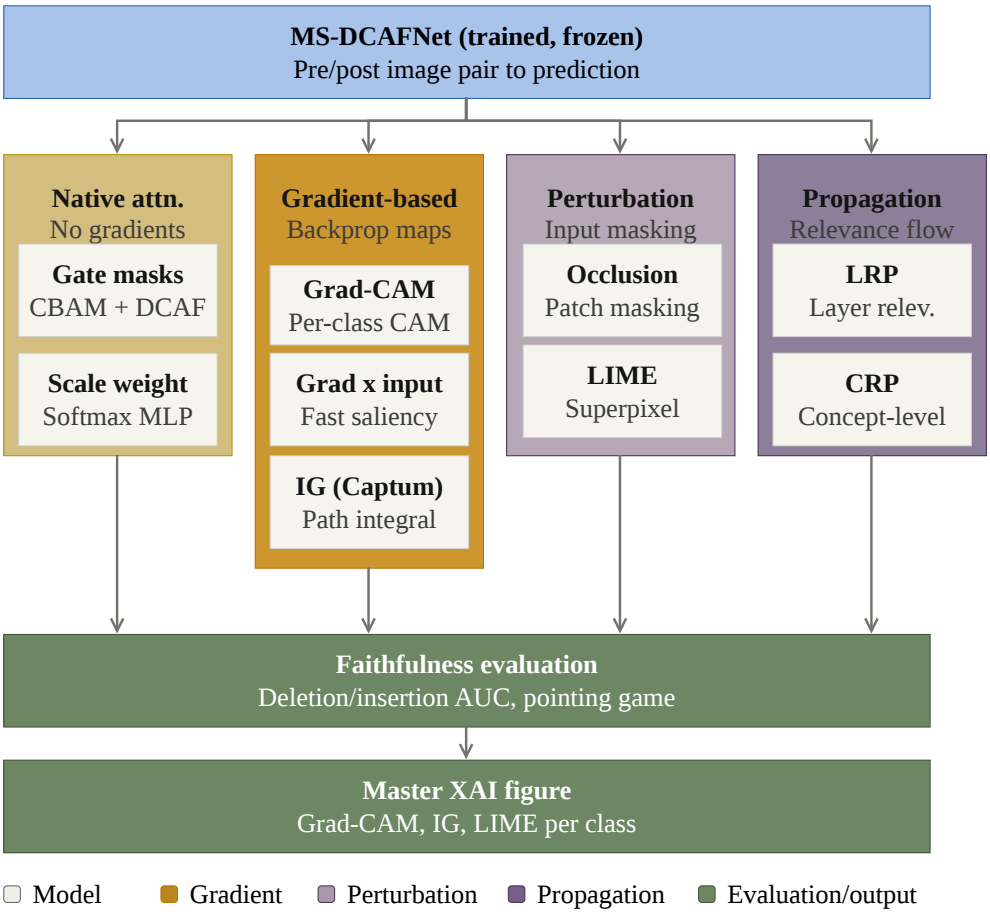


Figure 3.2: Overview of the explainability analysis pipeline: native architectural attention, gradient-based attribution, perturbation-based attribution, and relevance propagation methods are combined with quantitative faithfulness evaluation to produce the master XAI figure.

3.8.2 Gradient-Based Attribution

Three gradient-based methods attribute prediction confidence to input pixels via backpropagation, computed separately for the pre-disaster and post-disaster branches: bi-temporal Grad-CAM (computed at the pre-branch projection, post-branch projection, and the H-DCAF bottleneck aggregation output), Gradient$\times$Input (a fast, single-backward-pass saliency baseline), and Integrated Gradients (implemented via Captum with a zero-pixel baseline and 10 interpolation steps).

3.8.3 Perturbation-Based Attribution

Occlusion sensitivity systematically zeroes out $3 \times 32 \times 32$ patches with a stride of $(1, 8, 8)$ and records the resulting drop in class confidence. LIME fits a local linear surrogate model over SLIC superpixels (50 superpixels, 100 perturbation samples per image), producing separate

explanations for the pre- and post-disaster branches.

3.8.4 Relevance Propagation Methods

Layer-wise Relevance Propagation (LRP) is implemented using the Zennit library with the EpsilonPlusFlat composite, backpropagating class-specific relevance scores through every layer to the input pixels. Concept Relevance Propagation (CRP) extends this by conditioning relevance on individual channels at the H-DCAF bottleneck aggregation block, revealing which internal representations drive the model's attention.

3.8.5 Faithfulness Evaluation

Saliency maps are quantitatively evaluated using Deletion AUC (lower is better), Insertion AUC (higher is better), the Pointing Game hit rate (whether the maximum saliency location falls within the ground-truth damage region), and Pearson correlation between the saliency map and the ground-truth mask.

3.9 Evaluation Metrics

Model performance is reported using four complementary metrics. The primary metric is $F1_{damaged}$, the macro-averaged F1 score over the three active damage classes (Minor, Major, Destroyed), which directly measures the model's ability to discriminate between damage severities. Overall Accuracy (OA) and Cohen's Kappa (κ) are additionally reported, along with mean Intersection-over-Union (mIoU) averaged over all five classes. OA is acknowledged to be inflated by background pixel dominance, so $F1_{damaged}$ is used for all model comparisons and ablation rankings.

Chapter 4

Results and Analysis

4.1 Experimental Setup

All experiments use the xBD dataset under the strict event-based partitioning protocol described in Section 3.7. The official xBD training set (2,799 image triplets) was partitioned into 2,573 training samples across nine disaster events and 226 validation samples from one held-out event. The official xBD held-out test set (933 triplets) was reserved exclusively for final evaluation and never observed during training or validation. All experiments were conducted with a fixed random seed for reproducibility.

4.2 Final Test-Set Performance

Table 4.1 reports the aggregate test-set metrics of MS-DCAFNet at the best validation checkpoint (Epoch 76, validation $F1_{damaged} = 0.2025$).

Table 4.1: Aggregate test-set metrics of MS-DCAFNet on the official xBD held-out set (933 samples).

Metric	**Value**
$F1_{damaged}$ (primary)	0.3983
$F1_{macro}$ (all 5 classes)	0.5716
$F1_{fg}$ (foreground only)	0.4716
mIoU (all 5 classes)	0.4459
Overall Accuracy (OA)	0.9403
Cohen's Kappa (κ)	0.6117

The overall accuracy of 0.94 and Kappa of 0.61 (indicating substantial agreement beyond chance) reflect competent global classification. The primary metric, $F1_{damaged} = 0.3983$, reflects the harder challenge of discriminating between minority damage classes under severe class imbalance — a difficulty widely reported across the xBD benchmark literature.

4.3 Per-Class Analysis

Table 4.2 reports per-class precision, recall, F1-score, and IoU on the test set.

'Background' achieves near-perfect F1 (0.972) and IoU (0.945), confirming that the encoder effectively separates building footprints from non-building terrain. High recall across all

Table 4.2: Per-class test-set metrics of MS-DCAFNet.

Class	F1	Precision	Recall	IoU
Background	0.9716	0.9965	0.9479	0.9448
No Damage	0.6917	0.5884	0.8390	0.5287
Minor Damage	0.2811	0.1773	0.6790	0.1636
Major Damage	0.4404	0.3041	0.7983	0.2824
Destroyed	0.4733	0.3428	0.7640	0.3100
Macro avg.	**0.5716**	—	—	**0.4459**

damage classes (0.68–0.84) with comparatively lower precision reflects a deliberately conservative false-negative avoidance strategy enforced through class loss weighting, which prioritises recall for safety-critical damage classes over precision — in disaster response, missing a severely damaged building carries a greater human cost than a false alarm. 'Minor Damage' remains the most challenging class, as it is the least represented in the training distribution and its visual signature (subtle roof discolouration and minor surface cracking) is the hardest to distinguish from undamaged structures at 256×256 resolution.

4.4 Ablation Study

To isolate the contribution of the H-DCAF module, three ablation variants were trained, sharing the same encoder, decoder, loss, and training schedule, differing only in the temporal fusion strategy (Table 4.3).

Table 4.3: Ablation study comparing temporal fusion strategies.

Fusion Strategy	$F1_{damaged}$	Δ vs. ours
Subtraction (bottleneck)	0.1687	-0.0338
Concatenation (bottleneck)	0.1785	-0.0240
Single-scale DCAF (bottleneck)	0.1715	-0.0310
H-DCAF $\times$4 (ours)	**0.2025**	—

H-DCAF $\times 4$ scales outperforms all three baselines, with the largest gap against simple subtraction (-3.38pp), confirming that raw element-wise difference at a single scale cannot adequately represent multi-granular damage evidence. The single-scale DCAF variant underperforms the full model by 3.1pp, demonstrating that *where* DCAF is applied matters as much as the DCAF mechanism itself — applying it only at the coarsest bottleneck discards the fine-grained edge and texture cues present at finer scales that are essential for Minor and Major Damage discrimination.

4.5 Training Dynamics

Figure 4.1 shows the training and validation loss curves, the validation $F1_{damaged}$ trajectory, the mIoU/OA/Kappa evolution, per-class metrics at the best checkpoint, and the learning rate schedule over the 106 training epochs before early stopping.

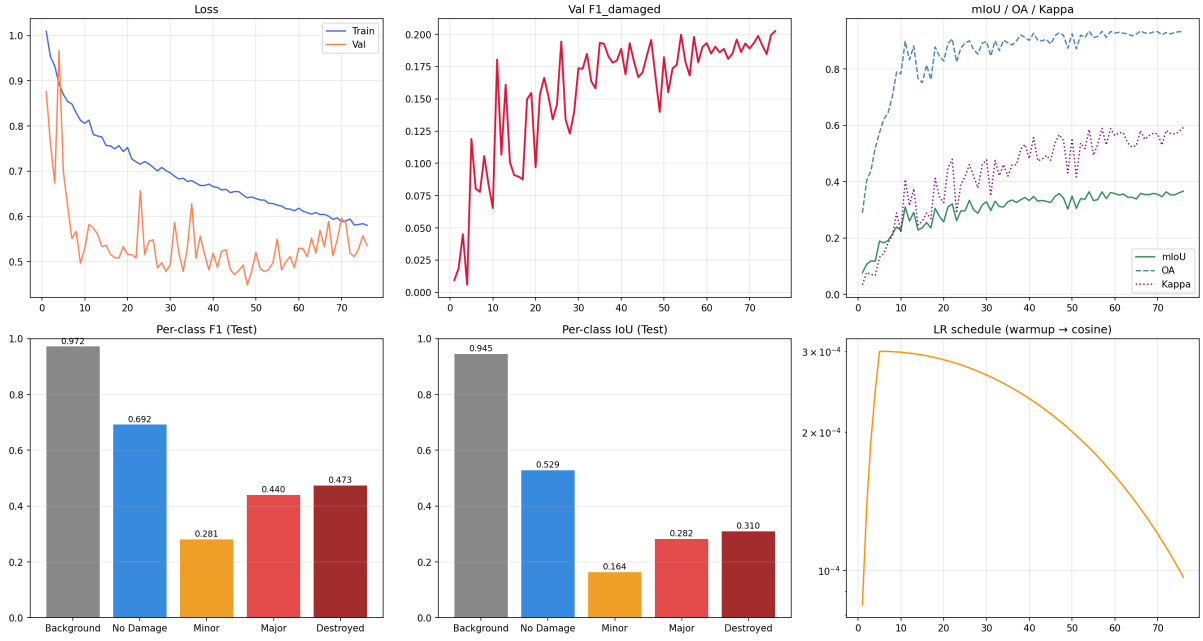


Figure 4.1: Training dynamics over 106 epochs: loss curves, validation $F1_{damaged}$ (peak 0.2025), mIoU/OA/Kappa trajectories, per-class F1/IoU at the best checkpoint, and the warmup-cosine learning rate schedule.

The train/validation loss curves converge in parallel without divergence, indicating that the event-based split successfully prevents spatial leakage — a model exploiting event-level memorisation would typically show rapidly decreasing training loss while validation loss plateaus or increases. The $F1_{damaged}$ curve shows characteristic noisy, monotonic improvement: high epoch-to-epoch variance is expected when learning under severe class imbalance, since small batch-to-batch variation in minority class composition produces large swings in the macro-F1 numerator.

4.6 Native Attention Analysis

Beyond post-hoc explanation, the architecturally intrinsic attention mechanisms of MS-DCAFNet provide direct interpretability without gradient computation overhead. Figure 4.2 visualises the H-DCAF scale weights, CBAM spatial attention maps, DCAF change-gate masks, and Haar band weights for a representative test sample.

The H-DCAF scale weights reveal that the model allocates the majority of its aggregation

weight to the two finest scales ($w_1 = 0.412$, $w_2 = 0.333$), with substantially lower weight at the two coarser scales ($w_3 = 0.120$, $w_4 = 0.135$) — consistent with a scene where fine building-boundary cues are more discriminative than block-level semantics. The CBAM spatial attention maps transition from broad, edge-driven suppression at Stage 1 to focused, object-level activation at Stage 4, reflecting the hierarchical abstraction expected in deep CNNs. The Haar band weights consistently favour the LH (horizontal edge) and HL (vertical edge) sub-bands over LL (smooth) across all active encoder stages, empirically confirming that the learnable wavelet decomposition prioritises edge and structural detail preservation over the low-frequency content that standard strided convolution would retain most strongly.

4.7 Post-Hoc Explainability Analysis

Figure 4.3 presents Grad-CAM saliency maps computed independently for each of the five damage classes across the pre-disaster branch, post-disaster branch, and fused bottleneck pathways.

Three patterns emerge. First, the pre-disaster pathway exhibits diffuse, scene-level activation, consistent with its role in establishing a pre-disaster structural reference. The post-disaster pathway shows sharper, building-localised activation, reflecting the shift toward post-disaster structural cues. Second, the fused bottleneck produces substantially more focused and spatially precise attributions than either individual pathway, confirming that H-DCAF synthesises complementary temporal information rather than simply averaging the two streams. Third, class-conditional specificity is evident: the response pattern for Major Damage concentrates on a distinct spatial region compared with Destroyed, demonstrating class-separable damage representations rather than a generic "something changed" detector.

Figure 4.4 presents Integrated Gradients attributions, computed independently for the pre- and post-disaster streams across the three active damage classes.

The IG maps highlight highly localised pixel clusters corresponding precisely to annotated building footprints, confirming that the model's class predictions are grounded in genuine structural changes rather than contextual or background cues. Consistent localisation across both Grad-CAM and IG — two methodologically independent attribution families — provides multi-method convergent evidence for the model's interpretability.

Figure 4.5 shows Layer-wise Relevance Propagation results backpropagated to the pre-disaster input pixels for all five damage classes.

The resulting per-class relevance maps corroborate the Grad-CAM and IG findings: high relevance is concentrated at annotated building locations across all five damage classes, with the destroyed-class map showing the most spatially compact and high-magnitude attribution — consistent with the clear, large-scale structural change signature of completely collapsed buildings.

Figure 4.6 consolidates all explanation modalities into a single per-class interpretability

dashboard for the Destroyed class.

Taken together, the post-hoc methods produce consistent localisation across all examined test samples, with attribution peaks aligning with annotated damage regions across Grad-CAM, IG, LRP, and LIME simultaneously. This multi-method convergence substantiates the claim that MS-DCAFNet's predictions are grounded in genuine structural damage evidence, providing the transparency required for operational trust in disaster-response decision support.

4.8 Pointing Game Evaluation

Quantitative faithfulness evaluation using the Pointing Game metric (Table 4.4) shows that Gradient×Input achieves the highest macro Pointing Game score (0.57), with particularly strong localisation for Major Damage (0.74), indicating reliable localisation of class-discriminative building regions. In contrast, the H-DCAF gate mask achieves a macro Pointing Game score of 0.07, consistent with its role as a change-sensitive, class-independent gating mechanism rather than a class-discriminative localisation module — it suppresses spatially unchanged regions before decoding rather than aligning with damage-class boundaries.

Table 4.4: Pointing Game hit rates on the xBD test set.

Method	**Minor** (n=15)	**Major** (n=39)	**Macro PG**
Gradient×Input	0.40	0.74	0.57
H-DCAF Gate (s_4)	0.13	0.00	0.07

4.9 Summary

The experimental evaluation demonstrates the effectiveness of the proposed MS-DCAFNet for multi-class building damage assessment from bi-temporal satellite imagery. The ablation study identifies H-DCAF as the primary contributor to performance gains over conventional temporal fusion strategies, while class-wise analysis indicates reliable performance across all damage categories, with Minor Damage remaining the most challenging due to its subtle visual characteristics and limited training representation. The interpretability analysis further confirms that the model focuses on meaningful structural damage cues, with consistent, convergent evidence from both native attention mechanisms and multiple post-hoc explanation methods.

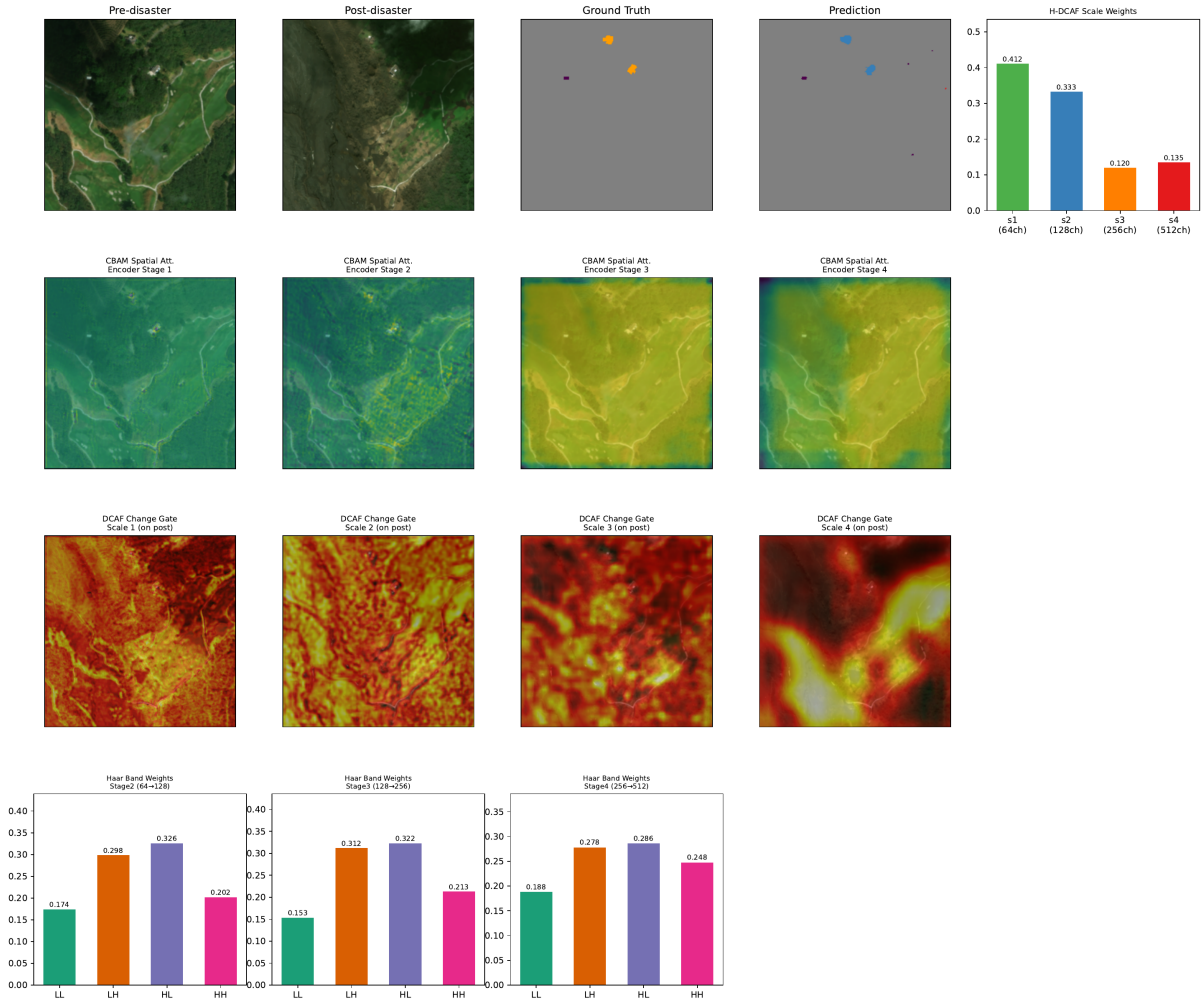


Figure 4.2: Native attention outputs: input pair, ground truth, and prediction (top row) alongside H-DCAF scale weights, CBAM spatial attention at all four encoder stages, DCAF change-gate masks, and Haar band weights at each downsampling stage.

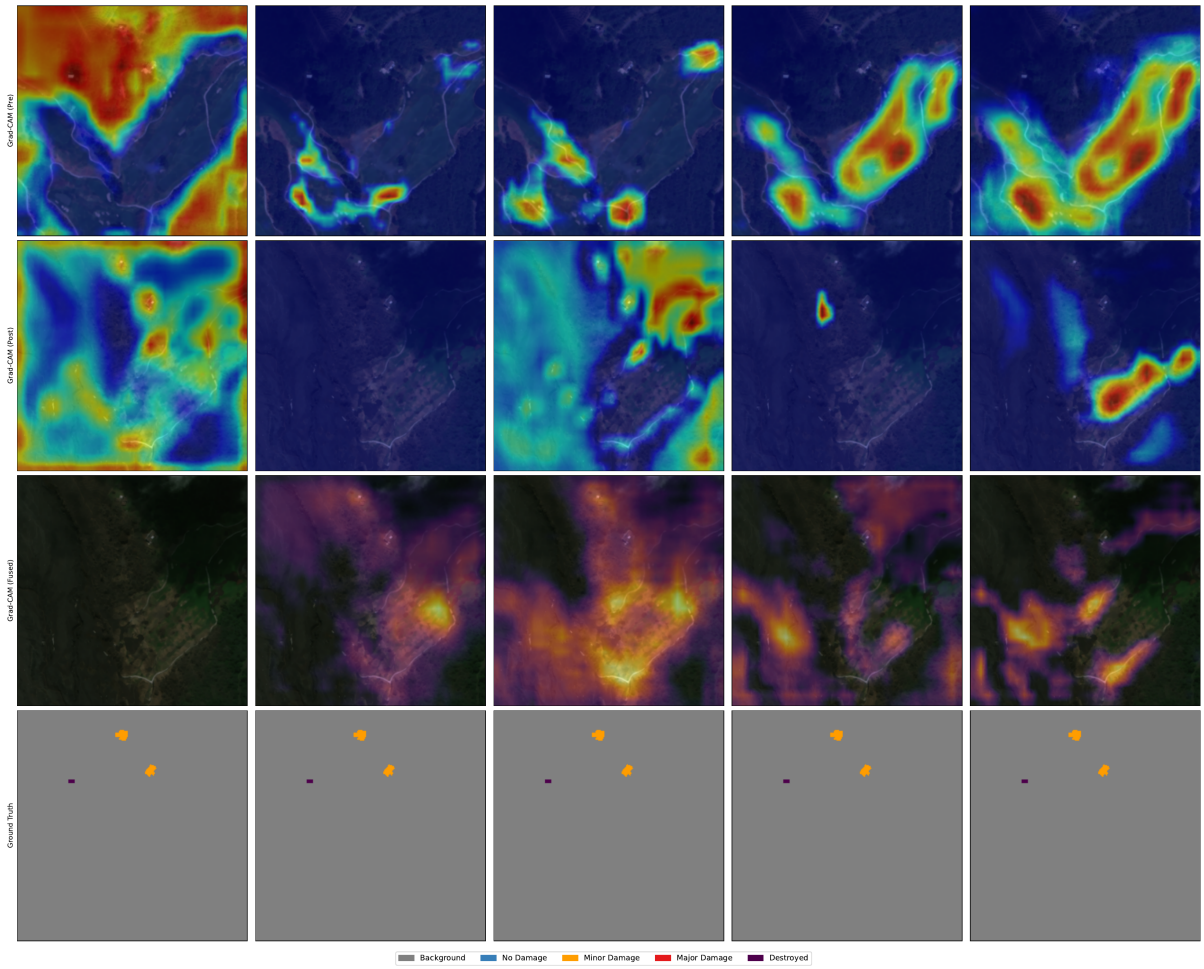


Figure 4.3: Grad-CAM saliency maps for all five damage classes across pre-disaster, post-disaster, fused bottleneck, and ground-truth rows.

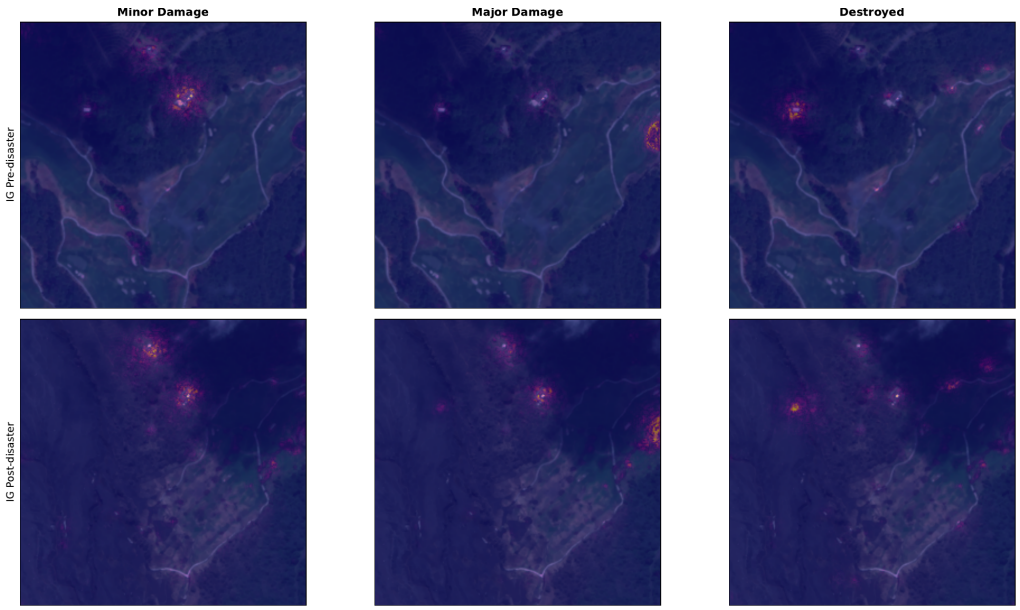


Figure 4.4: Integrated Gradients ($m = 10$ steps, zero baseline) for pre- and post-disaster branches across Minor, Major, and Destroyed classes.

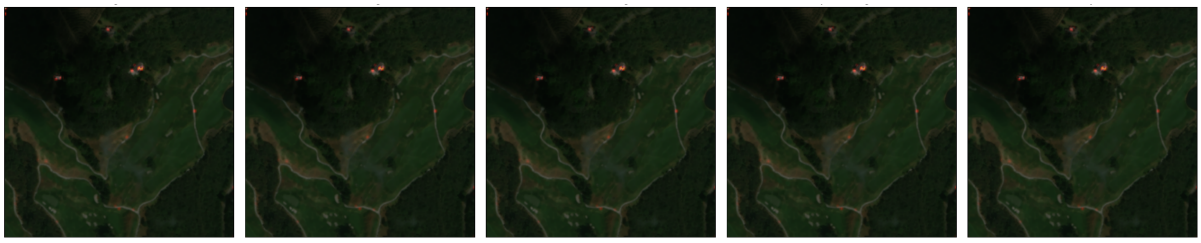


Figure 4.5: LRP (EpsilonPlusFlat) backpropagated to pre-disaster input pixels for all five damage classes. High relevance localises to annotated building footprints.

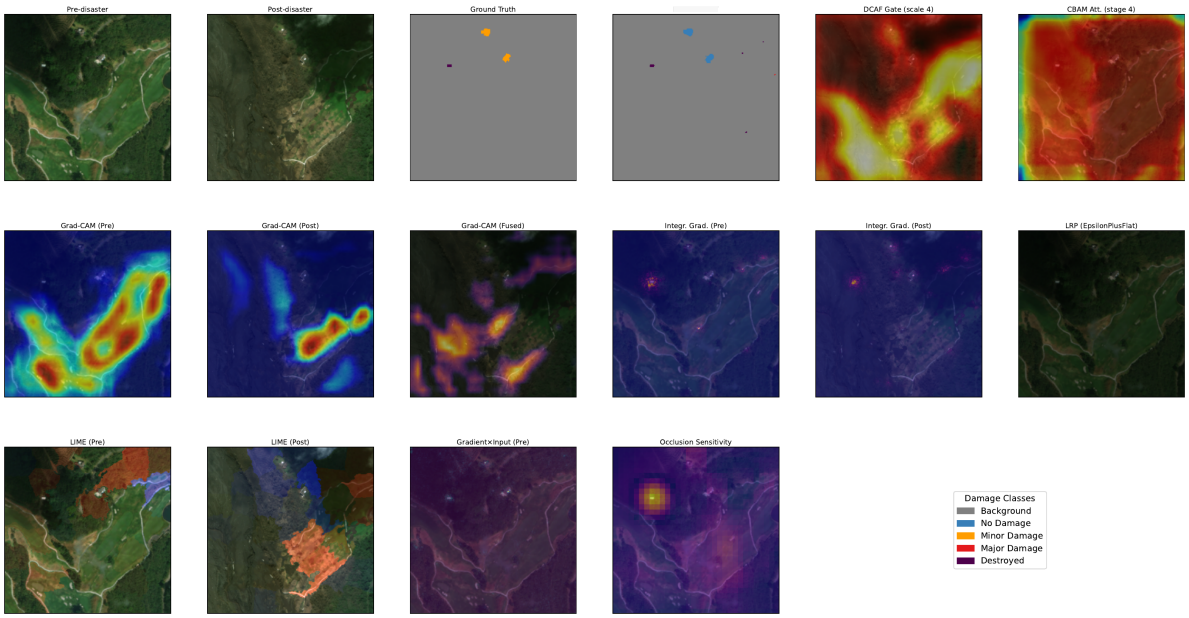


Figure 4.6: Comprehensive XAI panel for the Destroyed class, consolidating native attention, Grad-CAM, Integrated Gradients, LRP, LIME, Gradient×Input, and Occlusion Sensitivity. Attribution consistently localises to annotated building footprints across all modalities.

Chapter 5

Conclusion

5.1 Summary of Work

Building Damage Assessment (BDA) from bi-temporal high-resolution satellite imagery remains challenging due to information loss during feature extraction and the difficulty of modelling complex structural changes across time. During this internship, MS-DCAFNet was designed and implemented to directly address these two challenges: a wavelet-guided multi-scale Siamese encoder was built to preserve structural detail using learnable Haar wavelet decomposition in place of conventional downsampling, and a Hierarchical Damage-Conditioned Attention Fusion (H-DCAF) module was developed to adaptively fuse pre-disaster, post-disaster, and change-sensitive feature representations across multiple encoder scales, rather than relying on a single, fixed fusion point at the network bottleneck.

A UNet++-style decoder with deep supervision was implemented to exploit these fused multi-scale features for pixel-level damage localisation and classification, trained using a composite loss function combining focal, dice, and boundary-aware terms to handle the severe class imbalance present in the xBD dataset. The complete model was trained and evaluated under a strict event-based data partitioning protocol that prevents spatial leakage between training and test events — a stricter and more realistic evaluation setting than the standard xBD split used in much of the existing literature. The final model achieved an overall accuracy of 0.9403, a mean Intersection-over-Union of 0.4459, and a Cohen's Kappa of 0.6117, with controlled ablation experiments confirming that the proposed H-DCAF module consistently outperforms conventional temporal fusion strategies such as subtraction and concatenation.

Beyond predictive performance, a substantial part of the internship was devoted to building a comprehensive explainability (XAI) pipeline around the trained model, combining architecture-native attention, gradient-based attribution, perturbation-based methods, and relevance propagation, together with quantitative faithfulness evaluation. This analysis showed that the model's predictions are consistently grounded in meaningful structural damage cues rather than spurious contextual correlations, lending support to its reliability for operational disaster-response decision support.

5.2 Learning Outcomes

This internship provided hands-on experience across the full research and development lifecycle of a deep learning system, including:

- Designing and implementing a novel deep learning architecture from first principles, including custom wavelet-based downsampling and attention-based fusion modules in PyTorch.
- Working with large-scale, real-world remote sensing data, including the practical challenges of severe class imbalance, event-based data partitioning, and preventing spatial data leakage.
- Designing and tuning a composite loss function suited to a genuinely imbalanced, multi-class segmentation problem.
- Conducting rigorous, controlled ablation studies to isolate and validate the contribution of individual architectural components.
- Applying a broad range of explainable AI (XAI) techniques — including Grad-CAM, Integrated Gradients, LRP, CRP, and LIME — and critically interpreting their outputs to assess model trustworthiness.
- Preparing results and findings for academic publication, including manuscript writing, figure preparation, and adherence to IEEE formatting standards.

5.3 Limitations and Future Work

A few limitations of the current work were identified over the course of the internship. All experiments were conducted at a 256×256 input resolution, while several of the strongest published methods for this task operate at 512 px or higher; the performance gap associated with resolution is well documented in the segmentation literature and is a likely contributor to the comparatively low performance on the Minor Damage class in particular. Direct numerical comparison against other event-based methods was also limited, as different works adopt different event-based splits, precluding a fully controlled comparison.

Future work will extend MS-DCAFNet to higher input resolutions (512 px and above) and evaluate the model on a standardised, publicly available event-based benchmark split to enable direct, reproducible comparison against related methods. Further improvements to the Minor Damage class — through targeted data augmentation, resolution-aware architectural changes, or auxiliary supervision signals — are also identified as a priority direction.

5.4 Conclusion

Overall, this internship resulted in the design and development of MS-DCAFNet, an accurate, interpretable, and computationally efficient framework for automated multi-class building damage assessment from bi-temporal high-resolution satellite imagery. The work combines hierarchical temporal fusion, frequency-aware feature extraction, and comprehensive interpretability analysis, and the resulting model and findings are being prepared for submission to a peer-reviewed venue. The internship provided valuable exposure to applied deep learning research, remote sensing data challenges, and the practical demands of building trustworthy, explainable AI systems for safety-critical applications.

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